

## 8.2 Solving Linear Recurrence Relations

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Definition: A linear homogeneous recurrence relation is a recurrence relation of the form  $a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$  where  $c_1, \dots, c_k$  are constants and  $c_k \neq 0$ . } with constant coefficients

Linear because the right-hand side is only a sum of previous terms, no products ( $a_{n-1}a_{n-2}$  or  $(a_{n-3})^2$ )

Homogeneous because each term is a multiple of some  $a_i$ , no extra values.

The degree of such a recurrence relation is  $k$ , because  $a_n$  may be expressed using the previous  $k$  terms.

$$a_n = 2a_{n-2} + a_{n-3} \quad \text{Yes}$$

$$b_n = -nb_{n-2} \quad \text{Yes but not constant coefficient}$$

$$c_n = -(n+1)c_{n-1} + n-2 \quad \text{Not homogeneous}$$

$$d_n = (d_{n-1})^2 + 2d_{n-2} \quad \text{Not linear}$$

$$a_n \text{ is degree 3} \quad c_n \text{ is degree 1}$$

$$b_n \text{ is degree 2} \quad d_n \text{ is degree 2}$$

Definition: Given a recurrence relation  $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ , the associated equation  $r^k - c_1 r^{k-1} - \dots - c_k = 0$  is called the characteristic equation.

Solutions of this equation are the characteristic roots.

Theorem: Let  $c_1, c_2 \in \mathbb{R}$ . Suppose  $r^2 - c_1 r - c_2 = 0$  has distinct roots  $r_1, r_2$ . Then  $\{a_n\}$

is a solution of  $a_n = c_1 a_{n-1} + c_2 a_{n-2}$  if and only if  $a_n = \alpha_1 (r_1)^n + \alpha_2 (r_2)^n$  for  $n = 0, 1, 2, \dots$  and  $\alpha_1, \alpha_2$  some constants.

Find the solution to  $a_n = a_{n-1} + 2a_{n-2}$

if  $a_0 = 2$  and  $a_1 = 7$ .

$$\text{Characteristic Equation} \Rightarrow t^2 - t - 2 = 0$$

Characteristic Equation  $\Rightarrow t - t - 2 = 0$

So...  $(t-2)(t+1)=0$

$$a_n = \alpha_1 (-1)^n + \alpha_2 (2)^n \quad \boxed{t = -1, 2}$$

$$a_0 = 2 = \alpha_1 (-1)^0 + \alpha_2 (2)^0 = \alpha_1 + \alpha_2$$

$$a_1 = 7 = \alpha_1 (-1)^1 + \alpha_2 (2)^1 = -\alpha_1 + 2\alpha_2$$

$$\Rightarrow \begin{cases} \alpha_1 + \alpha_2 = 2 \\ -\alpha_1 + 2\alpha_2 = 7 \end{cases} \Rightarrow \begin{cases} \alpha_1 + \alpha_2 = 2 \\ 3\alpha_2 = 9 \end{cases}$$

$$\Rightarrow \alpha_2 = 3 \Rightarrow \alpha_1 + 3 = 2 \Rightarrow \alpha_1 = -1$$

$$\text{So } \{a_n\} = \{(-1)(-1)^n + 3(2)^n\}_{n=0}^{\infty}$$

Why do solutions look like this?

Suppose

$$a_n = c_1 a_{n-1} + c_2 a_{n-2} + \dots + c_k a_{n-k}$$

Can we let  $a_n = r^n$ ?

$$r^n = c_1 r^{n-1} + c_2 r^{n-2} + \dots + c_k r^{n-k}$$

$$\Leftrightarrow r^n - c_1 r^{n-1} - c_2 r^{n-2} + \dots + c_k r^{n-k} = 0$$

$$\Leftrightarrow (r^k - c_1 r^{k-1} - c_2 r^{k-2} + \dots + c_k) r^{n-k} = 0$$

so either  $r=0$  and  $\{a_n\} = \{0\}_{n=0}^{\infty}$

or  $r$  is a solution/root of

$$r^k - c_1 r^{k-1} - c_2 r^{k-2} + \dots + c_k$$

So  $a_n = r^n$  iff  $r$  is a

characteristic root

Why do we take linear combinations?

Suppose  $\{t_n\}_{n=0}^{\infty}$  and  $\{s_n\}_{n=0}^{\infty}$  are two

different solutions of the recurrence

relation  $a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$ .

Then if  $\alpha_1, \alpha_2 \in \mathbb{R}$ , is it true that

$\{\alpha_1 t_n + \alpha_2 s_n\}_{n=0}^{\infty}$  is also a solution?

$$\alpha_1 t_n + \alpha_2 s_n = \alpha_1 (c_1 t_{n-1} + \dots + c_k t_{n-k}) + \alpha_2 (c_1 s_{n-1} + \dots + c_k s_{n-k})$$

$$= c_1 \alpha_1 t_{n-1} + \dots + c_k \alpha_1 t_{n-k} + c_1 \alpha_2 s_{n-1} + \dots + c_k \alpha_2 s_{n-k}$$

$$= c_1 (\alpha_1 t_{n-1} + \alpha_2 s_{n-1}) + \dots + c_k (\alpha_1 t_{n-k} + \alpha_2 s_{n-k})$$

Yes! Any linear combination of solutions is also a solution.

Then if solutions look like  $a_n = r^n$

where  $r$  is a characteristic root,

then  $a_n = \alpha_1 (r_1)^n + \dots + \alpha_k (r_k)^n$

where each  $r_i$  is a characteristic root!

Find an explicit formula for the Fibonacci numbers.

$$f_n = f_{n-1} + f_{n-2}, \quad f_0 = 0, \quad f_1 = 1.$$

$$\Rightarrow r^2 - r - 1 = 0$$

$$\Rightarrow r = \frac{1 \pm \sqrt{1 - 4(-1)}}{2} = \frac{1 \pm \sqrt{5}}{2}$$

$$r_1 = \frac{1 + \sqrt{5}}{2}, \quad r_2 = \frac{1 - \sqrt{5}}{2}$$

$$\{f_n\} = \left\{ \alpha_1 \left( \frac{1 + \sqrt{5}}{2} \right)^n + \alpha_2 \left( \frac{1 - \sqrt{5}}{2} \right)^n \right\}_{n=0}^{\infty}$$

$$f_0 = 0 = \alpha_1 \left( \frac{1 + \sqrt{5}}{2} \right)^0 + \alpha_2 \left( \frac{1 - \sqrt{5}}{2} \right)^0 = \alpha_1 + \alpha_2$$

$$f_1 = 1 = \alpha_1 \left( \frac{1 + \sqrt{5}}{2} \right) + \alpha_2 \left( \frac{1 - \sqrt{5}}{2} \right)$$

$$\Rightarrow \begin{cases} \alpha_1 + \alpha_2 = 0 \\ \left( \frac{1 + \sqrt{5}}{2} \right) \alpha_1 + \left( \frac{1 - \sqrt{5}}{2} \right) \alpha_2 = 1 \end{cases} \quad \left( \alpha_1 = -\alpha_2 \right)$$

$$\Rightarrow - \left( \frac{1 + \sqrt{5}}{2} \right) \alpha_2 + \left( \frac{1 - \sqrt{5}}{2} \right) \alpha_2 = 1$$

$$\Rightarrow \frac{-1 - \sqrt{5} + 1 - \sqrt{5}}{2} \alpha_2 = 1$$

$$\Rightarrow \alpha_2 (-\sqrt{5}) = 1 \text{ or } \alpha_2 = -\frac{1}{\sqrt{5}} \Rightarrow \alpha_1 = \frac{1}{\sqrt{5}}$$

$$\text{So } f_n = \frac{1}{\sqrt{5}} \left( \frac{1 + \sqrt{5}}{2} \right)^n - \frac{1}{\sqrt{5}} \left( \frac{1 - \sqrt{5}}{2} \right)^n$$

↓  
This is called the golden ratio

Theorem: Let  $c_1, c_2 \in \mathbb{R}$  with  $c_2 \neq 0$ .

Suppose  $r^2 - c_1 r - c_2 = 0$  has only one

root (multiplicity 2)  $r_0$ . Then  $\{a_n\}$

is a solution of  $a_n = c_1 a_{n-1} + c_2 a_{n-2}$

if and only if  $\{a_n\} = \left\{ \alpha_1 (r_0)^n + \alpha_2 \cdot n \cdot (r_0)^n \right\}_{n=0}^{\infty}$

where  $\alpha_1, \alpha_2$  are constants.

Find the solution to  $a_n = 6a_{n-1} - 9a_{n-2}$  if  $a_0 = 1, a_1 = 6$ .

$$\Rightarrow t^2 - 6t + 9 = 0 \Rightarrow (t - 3)^2 = 0$$

$\Rightarrow t = 3$  is the only root (multiplicity 2)

$$\rightarrow t - 3t + 1 = 0 \Rightarrow (t-3) = 0$$

$\Rightarrow t=3$  is the only root (multiplicity 2)

$$\text{Then } a_n = \alpha_1(3)^n + \alpha_2 n(3)^n.$$

$$a_0 = 1 = \alpha_1(3)^0 + \alpha_2(0)(3)^0 = \alpha_1$$

$$a_1 = 6 = \alpha_1(3)^1 + \alpha_2(1)(3)^1 = 3\alpha_1 + 3\alpha_2$$

$$\Rightarrow \begin{cases} \alpha_1 = 1 \\ 3\alpha_1 + 3\alpha_2 = 6 \end{cases} \Rightarrow \begin{aligned} 3 + 3\alpha_2 &= 6 \\ \Rightarrow \alpha_2 &= 1 \end{aligned}$$

$$\text{So } \{a_n\} = \{(3)^n + n \cdot (3)^n\}_{n=0}^{\infty}$$

How do we verify that these solutions actually are solutions?

Plug it in

Theorem: Let  $c_1, \dots, c_k \in \mathbb{R}$ . Suppose  $r^k - c_1 r^{k-1} - \dots - c_k = 0$  has distinct roots  $r_1, \dots, r_k$ . Then  $\{a_n\}_{n=0}^{\infty}$  is a solution of the recurrence relation

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$$

if and only if

$$\{a_n\} = \{\alpha_1(r_1)^n + \dots + \alpha_k(r_k)^n\}_{n=0}^{\infty}$$

where  $\alpha_1, \dots, \alpha_k$  are constants.

Degree  $k$   
recurrence  
distinct  
roots

Theorem: Let  $c_1, \dots, c_k \in \mathbb{R}$ . Suppose

$$r^k - c_1 r^{k-1} - \dots - c_k = 0$$

has roots  $r_1$  with multiplicity  $m_1$ ,

$r_2$  with multiplicity  $m_2$ , ...

$r_s$  with multiplicity  $m_s$

( $m_1 + m_2 + \dots + m_s = k$ ,  $s \leq k$ ). Then  $\{a_n\}$  is a solution of

$$a_n = c_1 a_{n-1} + \dots + c_k a_{n-k}$$

if and only if

$$\begin{aligned} a_n &= (\alpha_{1,0} + \alpha_{1,1}n + \dots + \alpha_{1,m_1-1}n^{m_1-1})(r_1)^n \\ &+ (\alpha_{2,0} + \alpha_{2,1}n + \dots + \alpha_{2,m_2-1}n^{m_2-1})(r_2)^n \\ &+ \dots \\ &+ (\alpha_{s,0} + \alpha_{s,1}n + \dots + \alpha_{s,m_s-1}n^{m_s-1})(r_s)^n \end{aligned}$$

Degree  $k$   
recurrence  
multiple shared  
roots

If the roots of a recurrence relation are  $-1, 2, 2, 3, 3, 3, 8$ , then what is the general form of the solution?

$$-1 \rightarrow m=1 \quad 3 \rightarrow m=4$$

$$2 \rightarrow m=2 \quad 8 \rightarrow m=1$$

$$\{a_n\} = \left\{ \alpha_1 (-1)^n + (\alpha_{2,0} + \alpha_{2,1}n)(2)^n + (\alpha_{3,0} + \alpha_{3,1}n + \alpha_{3,2}n^2 + \alpha_{3,3}n^3)(3)^n + \alpha_4(8)^n \right\}_{n=0}^{\infty}$$

for some constants  $\alpha_1, \alpha_{2,0}, \alpha_{2,1}, \alpha_{3,0}, \alpha_{3,1}, \alpha_{3,2}, \alpha_{3,3}, \alpha_4$

Find the solution of  $a_n = 2a_{n-1} + a_{n-2} - 2a_{n-3}$

if  $a_0 = 3, a_1 = 6, a_2 = 0$ .

$$t^3 - 2t^2 - t + 2 = 0 \Rightarrow t^2(t-2) - (t-2) = 0$$

$$\Rightarrow (t^2 - 1)(t-2) = 0 \Rightarrow (t-1)(t+1)(t-2) = 0$$

$$t = 1, -1, 2$$

$$a_n = \alpha_1(1)^n + \alpha_2(-1)^n + \alpha_3(2)^n$$

$$a_n = \alpha_1 + \alpha_2(-1)^n + \alpha_3(2)^n$$

$$a_0 = 3 = \alpha_1 + \alpha_2 + \alpha_3$$

$$a_1 = 6 = \alpha_1 - \alpha_2 + 2\alpha_3$$

$$a_2 = 0 = \alpha_1 + \alpha_2 + 4\alpha_3$$

$$\Rightarrow \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 3 \\ \alpha_1 - \alpha_2 + 2\alpha_3 = 6 \\ \alpha_1 + \alpha_2 + 4\alpha_3 = 0 \end{cases} \Rightarrow \begin{cases} \alpha_1 + \alpha_2 + \alpha_3 = 3 \\ -2\alpha_2 + \alpha_3 = 3 \\ 3\alpha_3 = -3 \end{cases}$$

$$\Rightarrow \alpha_3 = -1 \Rightarrow -2\alpha_2 - 1 = 3 \Rightarrow \alpha_1 - 2 - 1 = 3$$

$$\Rightarrow -2\alpha_2 = 4 \Rightarrow \alpha_1 - 3 = 3$$

$$\Rightarrow \alpha_2 = -2 \Rightarrow \alpha_1 = 6$$

$$\{a_n\} = \left\{ 6 - 2(-1)^n - (2)^n \right\}_{n=0}^{\infty}$$